NITISH PRIYAN R S

Computer Science & Engineering Student | Software Engineer | Java Developer | AI/ML Engineer

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PROFESSIONAL SUMMARY

Computer Science undergraduate (Batch of 2028) and aspiring Software Engineer with strong proficiency in Java, backed by an Oracle Certified Associate Java SE 17 Developer credential and AWS Cloud Practitioner certification, with hands-on experience in Full-Stack Development, Machine Learning, Deep Learning, MLOps, and Natural Language Processing (NLP). Built and deployed 10+ projects spanning government interoperability platforms, medical image classification systems, and MLOps decision-support systems using Java, Python, REST APIs, FastAPI, React, Docker, and Git. Solved 170+ DSA problems in Java on LeetCode, with strong foundations in Object-Oriented Programming, Software Development Life Cycle (SDLC), and Agile practices.

TECHNICAL SKILLS

- Programming Languages: Java (Oracle Certified, SE 17), Python, C, C++, SQL
- Machine Learning & AI: Scikit-learn, PyTorch, XGBoost, MLflow, Gemini AI
- Web & Backend Development: FastAPI, REST API Design, React, Next.js, Flutter, Event-Driven Architecture
- Databases & DevOps: MySQL, Docker, Git, GitHub, CI/CD
- Core Concepts: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), MLOps, , API Integration, Problem Solving

PROJECT EXPERIENCE

SETU — Government Service Interoperability & Orchestration Platform | *Java, REST APIs, Event-Driven Architecture, RBAC* — GitHub

- Engineered a government interoperability and orchestration platform integrating fragmented departmental portals via REST API-based integration, common data models, consent-driven data sharing, event-driven workflows, and unified application tracking.
- Designed an AI-powered Service Guide, Life-Event Engine, and Service Graph to discover cross-department service dependencies, reuse verified citizen data/documents, and streamline multi-department application journeys, reducing manual coordination effort.
- Implemented SLA-risk analysis, bottleneck detection, interoperability monitoring, audit logging, Role-Based Access Control (RBAC), and automated notifications to improve cross-department coordination and service delivery visibility.

Diabetic Retinopathy Classification System | *Python, PyTorch, EfficientNet-B3, Grad-CAM, Next.js, Gradio* — GitHub

- Developed an end-to-end deep learning pipeline to classify Diabetic Retinopathy severity from retinal fundus images into 5 clinical levels, fine-tuning EfficientNet-B3 with class-weighted Focal Loss to handle class imbalance.
- Integrated Grad-CAM interpretability to visualize model decision regions, improving explainability for clinical review.
- Deployed the model via a Gradio backend on Hugging Face Spaces with a Next.js/React dashboard, achieving 71% overall validation accuracy across 5 severity classes.

Student Performance Prediction & MLOps Decision Support System | *Python, FastAPI, XGBoost, MLflow, DVC, Docker, CI/CD* — GitHub

- Built an end-to-end MLOps platform for student outcome prediction, risk scoring, and personalized intervention recommendations, achieving 89.1% prediction accuracy and 82.6% at-risk recall.
- Integrated MLflow for experiment tracking, DVC for data versioning, and Evidently AI for model/data drift monitoring within an automated CI/CD pipeline.
- Containerized the full pipeline with Docker and implemented asynchronous telemetry for real-time monitoring of model performance in production.

EDUCATION

Bannari Amman Institute of Technology — B.E. Computer Science and Engineering — Expected May 2028
CGPA: 7.79

Saratha Matriculation Higher Secondary School — HSC — 2023 – 2024
Score: 93.1%

ACHIEVEMENTS & ACTIVITIES

- Solved 170+ Data Structures and Algorithms (DSA) problems in Java on LeetCode, covering Linked Lists, Bit Manipulation, Arrays, Recursion, and core algorithmic topics.
- Selected participant, Smart India Hackathon (SIH) 2025 — national-level innovation and problem-solving competition.

CERTIFICATIONS

- Oracle Certified Associate: Java SE 17 Developer
- AWS Cloud Practitioner Essentials — Amazon Web Services (AWS)
- Introduction to Cybersecurity — Cisco Networking Academy

LANGUAGES & INTERESTS

- Languages Known: English (Professional Proficiency), Tamil (Native)
- Areas of Interest: Problem Solving, Software Development, Machine Learning, Building Projects